

The Verdict Is Not the Lever: An Interpretable Task-Completion Feature Predicts but Does Not Cause Long-Horizon Agent Termination

The mechanistic capstone of the WANDERING arc

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Abstract

A long-horizon coding agent fails in a distinctive way we call **WANDERING**: it keeps acting — reading, editing, running commands — but never emits the **finish** action, exhausting its turn budget. A four-paper arc detects WANDERING behaviorally and shows that residual steering cannot rescue it, while a content-free behavioral interruption can. Here we ask the deepest mechanistic version of the question: does the agent *internally represent* a “the task is done” verdict, and if so, is that representation the *causal lever* for termination? Using a full-stack sparse autoencoder (SAE) on Qwen3.6-27B over 99 SWE-bench Pro trajectories, we (i) isolate — anti-circularly, from SUCCESS trajectories only — an SAE feature at layer 23 (#22358) that is active at the final turn of WANDERING but not LOCKED runs (WANDERING-vs-LOCKED AUROC 0.81, length-controlled, partial- $r = 0.55$); (ii) interpret it *in-domain* as a “subtask completed and verified” feature (its top activations are statements like “all tests pass”, “the implementation is complete”; completion-language enrichment 50% vs. 6% baseline); (iii) show it *predicts the actual finish action* with AUROC 0.91 (ground-truth tool calls); yet (iv) **clamping it to the SUCCESS level at the WANDERING decision point produces no change in the probability of finishing** ($\Delta P(\text{finish}) = -0.001$, indistinguishable from a random-feature clamp at +0.002), and ablating it in SUCCESS does not significantly reduce finishing (-0.008 , n.s.). The agent represents “I’m done” as a clean, interpretable, predictive feature — *but that representation is not what makes it stop*. This extends the detection-versus-control asymmetry to the granularity of a single named feature, sharpening the arc’s three residual nulls: not the wrong direction or locus, but the right *interpretable* feature still failing as a lever. The only known rescue for WANDERING remains a behavioral interruption — the lever is behavioral, not representational. All experiments are CPU-only except a single forward pass for the causal test.

1 Background

WANDERING and the arc. Give a capable model a real software issue, a shell, and a fixed turn budget. Most runs end by submitting a patch via a **finish** tool (SUCCESS) or by locking onto a dead end and giving up early (LOCKED). A third, quieter failure is WANDERING: the agent keeps working and never decides it is done, exhausting the budget without ever calling **finish**.

A four-paper arc establishes a detection–intervention asymmetry around this mode. Paper #1 detects WANDERING from a residual-stream signature and a probe-free tool-entropy collapse, and proposes (its §13) a mechanism: *the mid-layer “verdict” that the task is done is consolidated, but the edge circuits that translate the verdict into the finish action fail to fire* [1]. Paper #2 pre-registers three residual interventions to supply the missing translation — injecting a SUCCESS-direction at the output layer and at the edge layer that best discriminates WANDERING — and all three are null on rescue [2]. Paper #3 localizes multi-channel signatures and finds that response to intervention is behavior-legible but residual-blind [3]. Paper #4, the first positive, shows that a transient *behavioral* interruption roughly doubles finalization where residual steering fails, and that the lever is the interruption itself, not its content [4]. A companion note shows the residual geometry of context-rot adds nothing predictive over the cheap behavioral detector [5].

The open mechanistic question. The arc’s §13 mechanism is, across all of it, only a *hypothesis*: “the verdict is consolidated but not emitted.” Two halves have never been tested directly. *First half*: is the “task is done” verdict an actual, interpretable internal object — a feature one can name — and is it present in WANDERING (which never terminates)? *Second half*: if it is present, is it the *cause* of termination, i.e. a lever? This paper answers both, using a sparse autoencoder to make the verdict a concrete feature and an activation clamp to test its causal role.

Why this is the right tool, and what is at stake. The broader interpretability field has repeatedly found that internal states which *predict* a behavior do not *control* it — a “knowledge–action gap” in which probes read errors at near-perfect AUROC while interventions correct almost none [6], and a “predict/control discrepancy” in which the features that predict a behavior are not those that steer it [7]. These results are established for single-forward-pass settings. What is unexamined is the asymmetry *on long-horizon agents*, at the level of a *single, interpretable, named feature* representing a specific decision (to terminate). If even the exact verdict feature is not a lever, the asymmetry is deeper than “we steered the wrong thing.”

2 Method

Data and SAE. The 99 Qwen3.6-27B SWE-bench Pro trajectories of the arc, labeled {SUCCESS:40, LOCKED:39, WANDERING:20}, with per-turn, per-position residual captures. A full-stack TopK SAE (caiovicentino1/qwen36-27b-sae-fullstack, $k = 128$, $d_{\text{sae}} = 40960$) at layer 23 (and 43 for replication) encodes the residual at the `pre_tool` position — the state immediately before the agent emits an action. Everything in Stages 0–2 runs on CPU on existing captures; the causal test (§3.4) is the only model forward pass. Each stage was pre-registered before the corresponding result was seen.

Stage 0 — does the verdict feature exist (first half of §13)? A candidate “done” feature is selected *from SUCCESS trajectories only*: features that fire more at the SUCCESS `finish`-turn than at SUCCESS early turns. WANDERING and LOCKED never touch the selection, so they are a held-out test of the prediction that WANDERING (which has reached completion) carries the verdict while LOCKED (which gave up) does not. We report per-trajectory early-window contrasts with a length confound control (partial- r given total trajectory length) and a within-trajectory trend test.

H1 — what is the feature (in-domain interpretation)? We read the transcript text at the turns where the selected feature fires hardest, and quantify completion/verification language against random and bottom-activation turns. (A general-corpus top-activation pass reads as boundary/punctuation tokens — a domain-mismatch artifact; the feature is agent-domain-specific, so the in-domain reading is decisive.)

E — does it predict the action (behavioral, ground truth)? Using the actual tool called at each turn (from the trajectory logs, no model), we measure the AUROC of the feature’s `pre_tool` activation for predicting whether the next action is `finish`.

H4 — is it the lever (causal)? We reconstruct the WANDERING and SUCCESS final-turn *decision point* by re-running the model on the original prompt (reusing the harness’s own prompt assembly so the reconstruction is faithful) up to the token where the tool name is chosen, and read $P(\text{finish})$ as the next-token mass over the three tools. We then *clamp* feature #22358 at layer 23 to the SUCCESS-final level at that position and re-read $P(\text{finish})$. Controls: a random *active* feature clamped to the same magnitude (null), and ablation of the feature in SUCCESS (necessity). Tests are paired per trajectory.

3 Results

3.1 Stage 0: the verdict feature exists and is held by WANDERING

A single SUCCESS-selected feature (#22358 at L23) carries the signal. It is active at the WANDERING final turn but not at the LOCKED final turn — exactly the §13 prediction that WANDERING holds a verdict LOCKED does not (Table 1). The effect survives the length control (partial- $r = 0.55$) and replicates, more weakly, at layer 43 (AUROC 0.60). With the program’s standard discipline: this is an uncorrected, single-layer-clean result — nothing survives multiple-comparison correction over the full metric×layer grid — so the claim rests on the convergence of Stages 0/H1/E/H4 below, not on one corrected p -value.

Table 1: Stage 0 (held-out WANDERING/LOCKED test of the SUCCESS-selected feature #22358). The feature is present at the WANDERING decision point and absent at LOCKED.

	L23 (primary)	L43 (replication)
WANDERING-final > early (P1)	$p = 2 \times 10^{-4}$	$p = 0.02$
WANDERING-final > LOCKED-final (P2)	$p = 1.7 \times 10^{-5}$, AUROC 0.81	$p = 0.01$, AUROC 0.60
survives length control (partial r)	0.55	0.30
top- K aggregate AUROC	0.89	0.77
mean act. S/W/W ₀ /L	1.01/0.58/0.00/0.15	1.81/0.32/0.00/0.05

3.2 H1: it is a “subtask completed and verified” feature

Reading the transcripts where #22358 fires hardest is unambiguous: the agent is confirming a unit of work. Top activations include “*All 157 tests pass. Let me view the final state...*”, “*The implementation is complete and tested...*”, “*The refactoring is complete. Let me provide a summary...*”, “*restoring the default caching behavior*”. Completion/verification language appears in 50% of the top-100 activating turns versus 6% of random turns and 5% of bottom turns; the `finish` tool is 18% of the top-60 activations versus 0.9% overall ($\sim 20\times$ enriched). The feature marks *a unit of work is complete*, firing at subtask completions throughout, not only at the global `finish`.

3.3 E: it predicts the finish action

Behaviorally, against ground-truth tool calls, #22358’s `pre_tool` activation predicts whether the next action is `finish` with **AUROC** 0.911 across all turns and 0.889 within SUCCESS turns alone

(a non-circular split, since SUCCESS has both `finish` and non-`finish` turns). Among the top decile of activation, the `finish`-rate is SUCCESS 12.2% vs. WANDERING 0.0% vs. LOCKED 0.0%: WANDERING reaches the high-completion state 84 times yet never converts (the never-converting is partly definitional; the load-bearing facts are that it forms the verdict repeatedly and the within-SUCCESS AUROC is 0.89). The feature is, by every observational measure, the “ready to finish” signal.

3.4 H4: clamping the verdict does not cause finishing

The reconstruction is behaviorally faithful: at the reconstructed decision point the model picks `finish` with probability 0.577 in SUCCESS and 0.075 in WANDERING — it recovers the real decision. At that point the feature itself reads 0.53 in SUCCESS and 0.00 in WANDERING, consistent with a collapse of the verdict by the moment of action (the structural profile shows it peaking before the action in SUCCESS and decaying in WANDERING). The causal test (Table 2) is a clean null. Clamping the verdict feature *up* to the SUCCESS level at the WANDERING decision point changes $P(\text{finish})$ by -0.001 — statistically and practically indistinguishable from a random-feature clamp ($+0.002$). Ablating the feature in SUCCESS trends in the necessity direction (-0.008) but is not significant: we cannot even establish that the feature is necessary for finishing. *The verdict is present, interpretable, and predictive — and it is not the lever.*

Table 2: H4 causal test (clamp at L23, WANDERING decision point; $n = 10/\text{class}$; paired Wilcoxon). The verdict-feature clamp is indistinguishable from a random-feature clamp.

condition	mean $\Delta P(\text{finish})$	median	p
clamp #22358 $\rightarrow$ SUCCESS level (0.53)	-0.001	$+0.000$	0.44
clamp random active feature (null)	$+0.002$	$+0.000$	0.94
ablate #22358 in SUCCESS ($\rightarrow 0$)	-0.008	$+0.000$	0.22

4 Discussion

Detection $\neq$ control, at the level of one named feature. The knowledge–action gap [6] and the predict/control discrepancy [7] are established for single-token errors. We show the same asymmetry for a *long-horizon behavioral decision* (to terminate), localized to a *single, interpretable, named* feature that represents exactly that decision. The agent forms a clean “I am done” representation, that representation predicts the `finish` action at AUROC 0.91, and yet supplying it where it has collapsed does nothing. This is one level deeper than the arc’s residual nulls [2]: those could be read as “we injected the wrong direction at the wrong layer.” Here the direction is the *right* one — selected to be the verdict, verified to be the verdict, shown to predict the action — and it is still inert.

What this says about §13. The mechanism’s first half is literally true: the verdict *is* consolidated as an interpretable feature, and WANDERING *does* carry it. But the second half — “the edge fails to emit” — is not a broken *readout from this feature*; restoring the feature does not drive emission. The decision to terminate is made downstream of, and not driven by, the representation that means “done.” WANDERING is best described not as “failing to form the verdict” but as “forming it and not treating it as terminal.”

The lever is behavioral. The only thing in the arc that rescues WANDERING is the transient behavioral interruption of paper #4 [4], which works without restoring any internal verdict. Taken

together: the predictive signal lives in the representation, the causal lever lives in behavior. For practitioners building agent reliability, this argues against “read the activations and nudge the right feature” and for cheap behavioral monitoring and behavioral course-correction.

5 Scope and honest caveats

Single model ($n=99$, Qwen3.6-27B), single task family (SWE-bench Pro). *Stage 0* is uncorrected/single-layer-clean (nothing survives FDR); the claim rests on convergence across stages, not one p -value. *H1*’s strict general-corpus semantic gate failed (boundary tokens) — the feature is domain-specific and read in-domain; a full per-token top-activation run would refine it. *H4* is the load-bearing negative and is modest in power: $n=10$ /class, a single layer and a single token position, behavioral-fidelity-gated (the cosine fidelity to the saved capture was not used), with prompts truncated to the last 4000 tokens for memory. The clean null (effect $\approx$ random null, far from significance) rules out *L23 #22358 as a single-position lever*; it does not prove that no verdict locus anywhere is a lever, and a multi-layer or multi-position clamp could in principle differ. We make no claim beyond: the right, interpretable, predictive verdict feature, clamped where it has collapsed, does not cause termination.

6 Conclusion

The WANDERING arc now closes mechanistically. *Detect* (tool-entropy collapse) $\rightarrow$ *localize* (edge layers) $\rightarrow$ *residual steering fails* (three nulls) $\rightarrow$ *behavioral interruption works* (first positive) $\rightarrow$ and now *the verdict is a real, interpretable, predictive feature that is still not the causal lever*. A long-horizon agent can represent “I have finished this” as cleanly as any concept we can name — and that representation is not what makes it stop. Code, pre-registrations, and data: github.com/OpenInterpretability/openinterp-swebench-harness (paper/verdict_circuit/).

References

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